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Systems and methods for real-time frequency band monitoring in a wireless network

Abstract

A User Equipment (“UE”) described herein may connect or request connection to a base station of a wireless network. The base station may implement multiple frequency bands, and connecting or requesting connection to the base station may include connecting or requesting connection via a particular frequency band of the multiple frequency bands. The UE may provide an indication of the particular frequency band to a core of the wireless network. The core may select a set of parameters for communications between the UE and the wireless network based on the indication of the particular frequency band. Additionally, or alternatively, the core may provide an indication of the particular frequency band to a device that provides a service to the UE via the wireless network, and the device may select parameters for the service based on the indication of the particular frequency band.

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Background/Summary

BACKGROUND

(1) Radio access networks (“RANs”) may serve as a wireless interface between User Equipment (“UEs”), such as mobile telephones, Internet of Things (“IoT”) devices, etc. and a core network

and/or other types of networks. RANs and UEs may operate according to different radio access technologies (“RATs”), such as a Long-Term Evolution (“LTE”) RAT, a Fifth Generation (“5G”) RAT, and/or other RATs. The various RATs may be associated with different frequency bands or sub-bands, such as a Sub-3 Gigahertz or “Sub-3” band, a C-Band (e.g., sometimes referred to as a “Sub-6” band), a millimeter-wave (“mmWave”) band, an Advanced Wireless Services (“AWS”) band, and/or other bands. Different frequency bands may deliver different speeds, provide for different connectivity range, and/or may have other differentiating attributes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a base station that implements multiple frequency bands;
- (2) FIG. 2 illustrates an example overview of one or more embodiments described herein;
- (3) FIG. 3 illustrates example communication interfaces via which a core network and/or other devices or systems may receive band information associated with a UE, in accordance with some embodiments;
- (4) FIGS. 4A and 4B illustrate an example of a device or system modifying service parameters based on band information received from a UE, in accordance with some embodiments;
- (5) FIG. 5 illustrates an example of requesting band information from a UE, in accordance with some embodiments;
- (6) FIG. 6 illustrates an example of selecting or modifying parameters of communications between a UE and a RAN and/or core network based on band information received from a UE, in accordance with some embodiments;
- (7) FIG. 7 illustrates an example process for providing or exposing band information from a UE **105** to a core network and/or other devices or systems, in accordance with some embodiments;
- (8) FIG. 8 illustrates an example environment in which one or more embodiments, described herein, may be implemented;
- (9) FIG. 9 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments;
- (10) FIG. 10 illustrates an example arrangement of an Open RAN (“O-RAN”) environment in which one or more embodiments, described herein, may be implemented; and
- (11) FIG. 11 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (12) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
- (13) As shown in FIG. 1, wireless network infrastructure of a RAN, such as a particular base station **101** (e.g., which may be, which may include, and/or may be communicatively coupled to an evolved Node B (“eNB”), a Next Generation Node B (“gNB”), a radio unit (“RU”), a remote RU (“RRU”), an Access Point (“AP”), etc.), may operate according to multiple different frequency bands **103** (referred to herein as “bands” **103** for the sake of brevity). In this example, assume that base station **101** operates according to bands **103-1**, **103-2**, and **103-3** (also referred to as “Band_1,” “Band_2,” and “Band_3”). For example, different bands **103** may each refer to a different RAT, band, sub-band, carrier, sub-carrier, and/or other portion of RF spectrum. Base station **101** may include, for example, hardware such as antennas, radios, etc. that are capable of operating according to bands **103-1** through **103-3**. In some embodiments, for example, bands **103-1** through **103-3** may include one or more LTE bands, 5G bands, or bands associated with other RATs.
- (14) One or more UEs **105** (e.g., mobile telephones, IoT devices, wearable devices, autonomous or

semi-autonomous wireless manufacturing robots, etc.) may connect to base station **101** via one or more of such bands **103**. For example, as shown, a particular UE **105** may communicate with base station **101** via band **103-2**. In some embodiments, different UEs **105** may be capable of communicating with base station **101** via some or all of bands **103-1** through **103-3**, and/or may be capable of wireless communications according to one or more other bands. In some embodiments, UE **105** may connect to base station **101** via a particular band **103** based on one or more different factors, such as signal quality or other RF metrics associated with bands **103** implemented by base station **101**, capabilities of UE **105** (e.g., where UE **105** may not necessarily support all bands **103** implemented by base station **101**), and/or other suitable factors. In some situations, UE **105** may switch bands after connecting to base station **101**, such as a switch from band **103-2** to band **103-1** or band **103-3**.

(15) UE **105** and/or base station **101** may maintain information indicating via which band or bands **103** UE **105** is connected to base station **101**. However, other devices, systems, networks, etc. may not be “aware” of the particular band **103** via which UE **105** is connected to base station **101**. For example, an application server or Multi-Access/Mobile Edge Computing (“MEC”) device, referred to sometimes herein simply as a “MEC,” that is providing a service to UE **105** may not have information indicating the particular band or bands **103** via which UE **105** is connected to base station **101**. In situations where different bands **103** provide different levels of performance (e.g., throughput, latency, etc.), the delivery of the service via different bands **103** may affect the overall performance, Quality of Service (“QoS”) metrics, and/or overall user experience. For example, if the service is relatively latency-sensitive and band **103-2** is congested or is otherwise unable to accommodate the latency needs of the service, the user experience associated with the service may be degraded, as excess latency may be introduced via band **103-2**. In accordance with some embodiments, the currently connected band may be provided by the UE to one or more other devices or systems, such that service parameters, network configuration parameters, and/or other parameters may be updated based on which band or bands **103** are being used by UE **105** to communicate with base station **101**. For example, the service in the example above may be provided at a lower data rate, and/or other parameters of the service may be modified to account for the added latency via band **103-2**. As discussed below, other elements of a network may perform other operations based on monitoring or receiving information indicating which band or bands **103** are used by UE **105** to communicate with base station **101**.

(16) For example, in accordance with some embodiments and as shown in FIG. 2, UE **105** may provide (at **102**) band information to a network access control system, such as Access and Mobility Management Function (“AMF”) **201**. While examples herein are described in the context of communications between UE **105** and AMF **201**, similar concepts may apply to communications between UE **105** and a Mobility Management Entity (“MME”) or other suitable element of a network, with which UE **105** is able to communicate. “Band information,” as referred to herein, may indicate which RAT, band, sub-band, carrier, sub-carrier, frequency, frequency range, etc. is associated with communications between UE **105** and base station **101** (e.g., via which band or bands UE **105** is currently connected to base station **101**).

(17) In some situations, UE **105** may be concurrently connected to base station **101** via multiple bands. In some such situations, one band may be an “active” band and one or more bands may be “idle” or “standby” bands. In some such situations, multiple bands may be “active” bands via which UE **105** actively sends and/or receives wireless traffic to and/or from base station **101**. The band information may, in some embodiments, indicate which band or bands **103** are “active” bands and which band or bands are “idle” or “standby” bands. An “active” band may include, for example, a band via which UE **105** actively sends and/or receives wireless traffic to and/or from base station **101**. For example, base station **101** may allocate RF resources for communications between UE **105** and base station **101** for an active band. An “idle” or “standby” band may include a band for which UE **105** and base station **101** maintain context information or other suitable

connection information, but via which UE **105** and base station **101** do not actively send or receive user plane traffic. For example, base station **101** may not allocate RF resources for communications between UE **105** and base station **101** for an idly or standby band.

(18) In some embodiments, UE **105** may provide the band information via Non-Access Stratum (“NAS”) signaling (e.g., via an N1 interface between UE **105** and AMF **201**). In some embodiments, UE **105** may provide the band information as part of an initial connection or attachment procedure with base station **101**. For example, AMF **201** may receive the band information via Registration Request message, a UE Capability Info Indication message, a measurement report (e.g., which may also include signal or channel quality measurements as measured by UE **105**), and/or some other suitable message from UE **105**. In some embodiments, UE **105** may provide the band information to AMF **201** after a connection to base station **101** has been established, such as when UE **101** has completed a UE-initiated or network-initiated switch from one band **103** to another. UE **105** may provide such band information to AMF **201** via a Radio Resource Control (“RRC”) Reconfiguration Complete message or other suitable message. In some embodiments, UE **105** may report band information to AMF **201** via some other suitable communication pathway, such as via one or more application programming interfaces (“APIs”). In some embodiments, UE **105** may execute an application, firmware, etc. that automatically reports the band information to AMF **201**. For example, UE **105** may report the band information to AMF **201** on a periodic basis (e.g., every minute, every hour, etc.), on an event-driven basis (e.g., based on connecting to base station **101**, based on switching bands **103** while connected to base station **101**, and/or based detecting the occurrence of on some other event), based on a request from AMF **201** or some other device or system, based on an input received at UE **105** (e.g., from a user of UE **105** or an application executing at UE **105**), and/or on some other suitable basis. In some embodiments, different UEs **105**, or types of UEs **105**, may report band information to AMF **201** in different manners. For example, a first type of UE **105** (e.g., IoT devices) may report band information on a relatively less frequent periodic basis than another type of UE **105** (e.g., mobile telephones). As another example, a first group of UEs **105** (e.g., a group of UEs **105** associated with a particular organization or category) may report band information on an event-driven basis (e.g., based on connecting to base station **101** or other events), while a second group of UEs **105** (e.g., UEs associated with a different organization or category) may report band information on a periodic basis.

(19) AMF **201** may provide the band information for UE **105** to one or more other devices or systems, such as RAN management system **203**, core network **205**, and/or application server **207**. For example, AMF **201** may provide (at **104**) band information for UE **105** to RAN management system **203**. RAN management system **203** may, for example, configure one or more parameters of base station **101**, such as queue weight parameters, beamforming parameters (e.g., azimuth angle, beam width, beam transmit power, etc. associated with one or more antennas or radios that implement bands **103**), RF allocation parameters, or other suitable RAN parameters. RAN management system **203** may modify (at **106**) one or more parameters of base station **101** and/or other base stations **101** based on receiving band information for UE **105** and/or one or more other UEs. For example, RAN management system **203** may determine that a particular band **103** is congested at a given location, and may “steer” one or more antennas associated with the particular band **103** away from the given location, in order to reduce congestion of the particular band **103** at base station **101**.

(20) AMF **105** may additionally, or alternatively, provide (at **108**) band information, for UE **105** and/or one or more other UEs, to one or more elements of core network **205**. Core network **205** may select or update (at **110**) routing information, policies, monitoring information, and/or other parameters based on the band information. For example, core network **205** may select or assign a particular network slice, a set of QoS parameters (e.g., traffic templates, QoS Class Identifier (“QCI”) values, 5G QoS Identifier (“5QI”) values, etc.), and/or other suitable parameters to

communications associated with UE **105** based on which particular band **103** is being used by UE **105** to communicate with base station **101**. For example, different network slices may be associated with different measures of QoS or other metrics, and core network **205** may select a particular network slice that, in combination with the currently connected band **103** associated with UE **105**, delivers at least a threshold measure of performance. As another example, core network **205** may maintain up-to-date usage information, which may be used by a policy and/or charging system of core network **205** to compare usage via the particular band **103** and a maximum allotment of usage via the particular band **103** associated with UE **105**.

(21) In some embodiments, one or more devices or systems external to core network **205** or a RAN that includes base station **101** may receive band information associated with UE **105**. For example, application server **207**, which may provide services to UE **105** (e.g., via core network **205** and/or one or more other networks), may receive (at **112**) an indication of the currently connected band **103** associated with UE **105**. Application server **207** may modify (at **114**) traffic and/or services provided to UE **105** based on the band information. For example, if UE **105** is connected to base station **101** via a relatively low throughput band, application server **207** may reduce a bitrate of traffic sent to UE **105**. If, on the other hand, UE **105** is connected to base station **101** via a relatively high throughput band, application server **207** may increase a bitrate of traffic sent to UE **105**. In practice, other operations may be performed by RAN management system **203**, core network **205**, application server **207**, and/or other suitable devices or systems based on receiving information indicating the particular band or bands **103** that is/are being used by UE **105** to communicate with base station **101**. Accordingly, the manner in which application server **207** provides services to UE **105** via base station **101** and/or core network **205** may take the currently connected band **103** into account, which may provide for an enhanced user experience as well as more efficient use of resources.

(22) FIG. 3 illustrates example interfaces between UE **105** and one or more other devices or systems, via which UE **105** may provide band information to such devices or systems. For example, as shown, UE **105** may provide band information to AMF **201** via an N1 interface, via NAS messaging, via control plane signaling, etc. In some embodiments, UE **105** may provide the band information to AMF **201** via base station **101**, such as by sending RRC messages to base station **101**, which may encapsulate, wrap, etc. such messages in a format, protocol, etc. associated with the N1 interface and/or otherwise associated with communications between base station **101** and AMF **201** (e.g., via one or more Next-Generation Application Protocol (“NGAP”) messages). The band information, as provided by UE **105**, may include one or more identifiers of UE **105**, such as a Subscription Permanent Identifier (“SUPI”), a Globally Unique Temporary Identifier (“GUTI”), an International Mobile Station Equipment Identity (“IMEI”) value, a Mobile Directory Number (“MDN”), and/or other suitable information based on which band information for a particular UE **105** may specifically associated with such particular UE **105**.

(23) AMF **201** may provide the band information for one or more UEs **105** to one or more other devices or systems, such as one or more other elements of core network **205**. AMF **201** may, for example, automatically provide the band information to the other devices or systems (e.g., periodically or intermittently, each time band information is received from a respective UE **105**, etc.), and/or may provide the band information to such devices or systems based on receiving a request from such devices or systems. For example, AMF **201** may provide the band information to a UE information repository of core network **205**, such as Unified Data Management function (“UDM”) **301** via an N8 interface or some other suitable communication pathway. In some embodiments, one or more other devices or systems that communicate with UDM **301** may request, receive, etc. the band information. For example, a charging or usage monitoring system may maintain information indicating which band or bands **103** have been used by UE **105**, and an account, subscription status, usage allotment, etc. associated with UE **105** may be updated accordingly.

(24) In some embodiments, AMF **201** and/or some other device or system of core network **205** (e.g., UDM **301**) may provide the band information to a device or system that serves an interface between core network **205** and one or more external devices or systems, such as Network Exposure Function (“NEF”) **303** (e.g., via a Nnef interface), a Service Capability Exposure Function (“SCEF”), or other suitable device or system.

(25) NEF **303** may accordingly allow devices external to core network **205**, such as application server **207** and/or other devices or systems, to receive band information for respective UEs **105**. For example, NEF **303** may maintain information indicating which devices or systems are authorized to receive band information for particular UEs **105**. In this example, NEF **303** may maintain information indicating that a particular application server **207** is authorized to receive band information for a particular UE **105**, and may accordingly provide such band information to application server **207** (e.g., may automatically “push” the band information to application server **207** or may respond with band information based on requests from application server **207**). While some examples of devices or systems that may receive band information are provided above, in practice, other devices or systems (e.g., other elements of core network **205**) may receive band information for one or more UEs **105** (e.g., from AMF **201**, UDM **301**, etc., via interfaces between such elements).

(26) FIGS. **4A**, **4B**, and **5** illustrate examples in which a device external to core network **205** (i.e., application server **207**, in these examples) receives band information for a particular UE **105**, and modifies service parameters based on the band information. As shown in FIG. **4A**, AMF **201** may receive, monitor, etc. (at **402** and **404**) band information associated with multiple UEs **105**, including UEs **105-1** and **105-2**. For example, as discussed above, UEs **105-1** and **105-2** may each push the band information periodically, on an event-driven basis, etc. to AMF **201** via NAS messaging while connected to a respective base station **101**. UEs **105-1** and **105-2** may provide (at **402** and **404**) the band information to AMF **201** in an asynchronous and/or independent manner, inasmuch as UE **105-1** provides (at **402**) respective band information to AMF **201** independently from UE **105-2** providing (at **404**) its own respective band information, and vice versa. In some embodiments, UEs **105-1** and **105-2** may provide (at **402** and **404**) band information in a synchronized manner, such as at the same periodicity. In some embodiments, UEs **105** may include band information when outputting connection requests to connect or attach to base station **101**. For example, band information may be included in an RRC Setup Request message, a NAS Registration Request, a Packet Data Network (“PDN”) Connectivity Request, or other suitable messaging between UE **105** and base station **101** and/or AMF **201**. In some embodiments, the band information may be included, or “piggybacked,” in one or more existing messages that are sent by UE **105** to base station **101** and/or AMF **201**.

(27) Based on receiving (at **402** and **404**) the band information, AMF **201** may store, maintain, etc. (at **406**) up-to-date band information for UEs **105-1** and **105-2**. For example, in situations where UE **105-1** and/or UE **105-2** were previously connected to a different band **103**, AMF **201** may update, overwrite, etc. previous band information to reflect the band or bands **103** to which UEs **105-1** and **105-2** are currently connected (e.g., as received at **402** and/or **404**). Thus, in the example shown in FIG. **4A**, AMF **201** may store, maintain, etc. (at **406**) up-to-date band information associated with UEs **105**. In some embodiments, AMF **201** may output band information to UDM **301** (e.g., via the N8 interface as discussed above or via some other suitable communication pathway), based on which UDM **301** may maintain band information for one or more UEs, such as UEs **105-1** and **105-2**.

(28) At some point, application server **207** may request (at **408**) band information for UE **105-2**. Application server **207** may be associated with UE **105-2**, such as by actively providing a service to UE **105-2** (e.g., communicating with UE **105-2** via core network **205** and/or some other network), by preparing to provide a service to UE **105-2** (e.g., based on a predicted or estimated demand for service, based on a scheduled event, etc.), and/or otherwise maintaining some association with UE

105-2. In some embodiments, application server **207** and/or UE **105-2** may have previously registered with NEF **303** and/or some other device or system indicating the association between UE **105-2** and application server **207**, and/or by otherwise indicating authorization for application server **207** to request and/or receive band information associated with UE **105-2**. In some embodiments, application server **207** may “subscribe” to or otherwise request band information associated with UE **105-2**, such as by providing one or more identifiers of UE **105-2** (e.g., an Internet Protocol (“IP”) address, an MDN, etc.), based on which NEF **303** is able to identify the authorization of application server **207** to receive band information associated with UE **105-2**. The request (at **408**) may accordingly include an identifier of UE **105-2**, an identifier and/or authentication information associated with application server **207**, and/or other suitable information based on which NEF **303** is able to determine that application server **207** is authorized to receive band information for UE **105-2**.

(29) Assuming NEF **303** has authenticated the request (at **408**) and/or has verified authorization for application server **207** to receive band information for UE **105-2**, NEF **303** may forward (at **410**) the request for band information, associated with UE **105-2**, to AMF **201**. AMF **201** may identify (at **412**) previously stored band information for UE **105-2**, and may respond (at **414**) with an indication of the currently connected band or bands **103** associated with UE **105-2**. As discussed above, AMF **201** may indicate which band or bands **103** is/are active bands and/or which band or bands **103** is/are idle or standby bands.

(30) Additionally, or alternatively, NEF **303** may request band information from one or more other sources, such as UDM **301**. As discussed above, UDM **301** may maintain band information and may accordingly respond to requests from NEF **303** or other authorized entities for such information. In some situations, UDM **301** may not maintain band information for one or more UEs **105**, such as situations in which UDM **301** has not yet received band information (e.g., from AMF **201**), or is not configured to receive or maintain band information. In some embodiments, based on determining that UDM **301** does not maintain the requested band information, UDM **301** may respond to NEF **303** with an indication that UDM **301** does not maintain such information. In such scenarios, NEF **303** may then request (at **410**) band information from AMF **201** and/or from some other source. Additionally, or alternatively, based on determining that UDM **301** does not maintain the requested band information, UDM **301** may request the band information from AMF **201** and may forward such information to NEF **303**.

(31) As another example situation, UDM **301** may maintain band information, along with a timestamp or some other temporal information associated with the band information (e.g., an indication of when the band information was received, generated, etc.). In some scenarios, UDM **301** may receive a request (e.g., from NEF **303**) for band information, and may provide the band information along with the associated temporal information. NEF **303** may, in some embodiments, determine whether the band information is “old,” stale,” etc. For example, if the band information was received, generated, etc. at least a threshold amount of time before NEF **303** outputted the request or received the band information, NEF **303** may determine that the band information is “old,” “stale,” etc. (referred to herein simply as “stale” for the sake of brevity), and may subsequently request (at **410**) the band information from AMF **201**. Additionally, or alternatively, UDM **301** may make a determination as to whether band information is stale. In situations where UDM **301** has determined that band information is stale, UDM **301** may request band information from AMF **201**, and provide such information to NEF **303**. Additionally, or alternatively, in situations where UDM **301** has determined that band information is stale, UDM **301** may indicate that UDM **301** does not have the requested band information, and/or that UDM **301** has stale band information, based on which NEF **303** may request (at **410**) the band information from AMF **201** or some other suitable source.

(32) Once NEF **303** has received (at **414**) the requested band information, NEF **303** may forward (at **416**) the band information for UE **105-2** to application server **207**. Application server **207** may,

in some situations, modify or select (at **418**) particular service parameters associated with one or more services, traffic, etc. provided to UE **105-2** (e.g., via core network **205**) based on the received band information. For example, as discussed above, application server **207** may modify a bitrate of traffic sent to UE **105-2** based on the band information, may change protocols or other service attributes based on band information, may select or de-select UE **105-2** as a host for a group messaging or gaming session, and/or may perform other suitable service parameter modifications based on the band information. Generally, as discussed above, application server **207** may make such modifications in order to account for differing performance attributes or other attributes associated with different bands **103**. Application server **207** may proceed to provide (at **420**) one or more services to UE **105-2** (e.g., via core network **205** and base station **101**) according to the modified or selected service parameters.

(33) In some embodiments, AMF **201** may provide (at **414**) band information in a request-response fashion. For example, AMF **201** may forgo providing such band information (e.g., to NEF **303** and/or application server **207**) in the absence of a request (e.g., at **410**), and may provide the band information after receiving such a request. In some embodiments, AMF **201** may provide (at **414**) band information on an periodic basis, on an event-driven basis (e.g., each time updated band information is received for UE **105-2**), and/or on some other basis. In this sense, application server **207** may “subscribe” to updates for band information associated with UE **105-2**, without needing to issue multiple requests for such information.

(34) As shown in FIG. **4B**, NEF **303** and/or some other device or system (e.g., other than AMF **201**) may maintain band information associated with UEs **105**. For example, as shown, AMF **201** may forward (at **405**) band information received (at **402** and **404**) from UEs **105-1** and **105-2** to NEF **303**. For example, AMF **201** may forward (at **405**) band information upon receiving (at **402** and **404**) such information. NEF **303** may maintain (at **407**) such band information at a local storage device, a remote cloud storage system, etc. When receiving (e.g., at **408**) a request for band information associated with UE **105-2**, NEF **303** may identify (at **409**) previously received band information for UE **105-2**, and may provide (at **411**) the band information for UE **105-2** to application server **207**. As similarly discussed above, application server **207** may modify or select (at **418**) particular service parameters based on the band information, and provide (at **420**) one or more services to UE **105-2** based on the modified or selected service parameters.

(35) As shown in FIG. **5**, AMF **201** may request, or “pull,” band information from UEs **105** (e.g., in lieu of, or in addition to, band information being “pushed” from UEs **105**). In this example, application server **207** may request (at **502**) band information associated with UE **105-2**. For example, application server **207** may request such information from NEF **303**. NEF **303** may determine that NEF **303** does not store band information for UE **105-2**. For example, in the example of FIG. **4B**, NEF **303** may be configured to store band information. However, situations may arise in which NEF **303** has yet to receive band information associated with UE **105-2**, or where band information stored by NEF **303** is “stale” (e.g., has been received at least a threshold time before the request is received). Additionally, or alternatively, in some embodiments, NEF **303** may not be configured to store band information for UE **105-2**.

(36) Based on NEF **303** not storing band information for UE **105-2** (e.g., including embodiments in which NEF **303** is not configured to store band information), NEF **303** may forward (at **504**) the request to AMF **201**. In a similar manner, if AMF **201** does not store the requested band information (e.g., in situations where AMF **201** is configured to maintain band information but has not yet received band information, where AMF **201** stores stale band information for UE **105-2**, and/or where AMF **201** is not configured to store band information), AMF **201** may request (at **506**) band information from UE **105-2**. For example, AMF **201** may output one or more messages via NAS signaling, via an N1 interface, via an API, etc. to UE **105-2**. In some embodiments, the message may include a Session Establishment Request, a Session Modification Request, a protocol data unit (“PDU”) Session Modification Command, and/or some other suitable message. In some

embodiments, AMF **201** may provide the request to base station **101**, which may wrap, encapsulate, translate, etc. the request to some other format, such as a RRC Reconfiguration Request that includes a flag, indicator, custom field, etc. that indicates that an appropriate response to the RRC Reconfiguration Request includes band information from UE **105-2**. While the message may be of a type that is typically used to modify parameters of a communication session associated with UE **105-2**, the message may, in some embodiments, not include any actual modifications to such parameters, and may instead be a technique for providing the band information request to UE **105**.

(37) UE **105-2** may respond (at **508**) with information indicating the particular band or bands **103** used by UE **105-2** to wirelessly communicate with base station **101**. For example, UE **105-2** may provide (at **508**) the band information via NAS signaling, an N1 interface, an API, etc. The band information may be provided via an RRC Reconfiguration Complete message, a PDU Session Establishment Accept message, or other suitable message. As similarly noted above, while the message may be of a type that is typically used to indicate the confirmation or acceptance of modified parameters of a communication session associated with UE **105-2**, the message may, in some embodiments, not indicate or be based on any actual modifications to such parameters, and may instead be a technique for providing the band information to AMF **201**.

(38) In some embodiments, AMF **201** may store and/or maintain (e.g., as discussed above with respect to operation **406**) the received (at **508**) band information. Additionally, or alternatively, AMF **201** may forward (at **510**) the band information to NEF **303**. Similarly, NEF **303** may, in some embodiments, store and/or maintain (e.g., as discussed above with respect to operation **409**) the received (at **510**) band information. NEF **303** may forward (at **512**) the requested band information to application server **207**. As similarly discussed above, application server **207** may select or modify (at **418**) service parameters, and provide (at **420**) one or more services to UE **105-2** based on the selected or modified service parameters.

(39) In some embodiments, as discussed above, core network **205** and/or a RAN (e.g., including base station **101**) may select or modify parameters based on the currently connected band associated with a given UE **105**. For example, as shown in FIG. **6**, AMF **201** may receive (at **602**), from Policy Control Function (“PCF”) **601** and/or some other suitable rule and/or policy system, a set of policies **603** relating to UE Route Selection Policy (“URSP”) rules or other suitable types of information. URSP rules may, for example, specify QoS rules, routing rules, and/or other information indicating a manner in which UE **105** should prioritize, queue, output, and/or perform other operations with respect to network traffic. For example, a particular URSP rule may specify that traffic having particular attributes or characteristics (also referred to as “traffic descriptors”) should be associated with a particular network slice. The URSP rule policy or policies **603**, provided (at **602**) by PCF **601** to AMF **201**, may include multiple different sets of URSP rules based on different bands **103**.

(40) For example, if UE **105** is connected to base station **101** via a first band **103-1**, then the URSP rule policy may specify a first set of URSP rules to be used by UE **105**, and if UE **105** is connected to base station **101** via a second band **103-2**, then the URSP rule policy may specify a second set of URSP rules to be used by UE **105**. For example, if UE **105** is connected to a first band **103-1** that is associated with relatively higher latency, then the URSP rules may specify that traffic matching a particular set of traffic descriptors should be associated with a highly prioritized network slice, which may offset or compensate for the relatively higher latency associated with band **103-1**. If, on the other hand, UE **105** is connected to a second band **103-2** that is associated with relatively lower latency, then the URSP rules may specify that traffic matching the same set of traffic descriptors should be associated with a less highly prioritized network slice, which may ultimately still yield at least a threshold measure of performance without consuming resources of the more highly prioritized network slice.

(41) FIG. **6** illustrates a simplified set of URSP rule policies **603** for the sake of explanation. For

example, as shown, URSP rule policies **603** may indicate that when UE **105** is connected via band **103-1**, voice traffic should be associated with a first network slice (“Slice 1”). Further, URSP rule policies **603** may indicate that when UE **105** is connected via band **103-2**, voice traffic should be associated with a second network slice (“Slice 2”).

(42) In some embodiments, PCF **601** may “push” the URSP rule policies to AMF **201**, such that AMF **201** stores and/or maintains URSP rule policies for one or more UEs **105**. Additionally, or alternatively, AMF **201** may request the URSP rule policies from PCF **601** during a registration or connectivity establishment procedure with UE **105**, a PDU session establishment procedure, or other suitable procedure during which AMF **201** communicates with UE **105**.

(43) As further shown, UE **105** may provide (at **604**) band information to AMF **201** (e.g., via NAS signaling, via an N1 interface, etc.), indicating a particular band or bands **103** via which UE **105** is connected to base station **101**. As discussed above, UE **105** may provide such information periodically, as part of a connection or communication session establishment request, etc. AMF **201** may identify a particular set of URSP rules that are associated with the indicated band or bands **103** (e.g., as indicated by URSP rule policies **603**), and may provide (at **606**) the relevant URSP rules (e.g., associated with the indicated band or bands **103**) to UE **105**. Additionally, or alternatively, AMF **201** may forward some or all of the URSP rule policies **603** to UE **105**, without first receiving (at **604**) band information from UE **105**.

(44) As further shown, UE **105**, base station **101**, and/or core network **205** may communicate (at **608**) in accordance with the URSP rules that were identified based on the currently connected band or bands **103** associated with UE **105**. For example, UE **105** may request a particular network slice for a particular communication session (e.g., matching traffic descriptors associated with one or more relevant URSP rules), which may include requesting such network slice for a new communication session or for the modification of an existing communication session.

(45) In some embodiments, application server **207** (e.g., which is in communication with UE **105**) may also have access to some or all of the URSP rule policies **603** provided to UE **105**. For example, application server **207** may receive (at **610**) URSP rule policies **603** from PCF **601** (e.g., via NEF **303**). As discussed above, NEF **303** may, for example, authenticate and/or determine that application server **207** is authorized to receive such information, which may be based on a registration procedure associated with UE **105** and/or application server **207**, and/or based on some other suitable procedure. In some embodiments, application server **207** may request URSP rule policies **603** from NEF **303**, and/or NEF **303** may “push” URSP rule policies **603** to application server **207** (e.g., based on an association between application server **207** and UE **105**). In this manner, application server **207** may have access to both band information associated with UE **105** (e.g., as received at **112**, **411**, **416**, **502**, etc.), as well as URSP rule policies **603** indicating particular URSP rules associated with particular bands. In this manner, application server **207** may be able to modify, configure, etc. services provided to UE **105** (e.g., modify a bitrate, modify a polling rate, modify QoS parameters, etc.) in order to conform with or otherwise account for URSP rules that are associated with the currently connected band for UE **105**.

(46) FIG. 7 illustrates an example process **700** for providing, exposing, etc. band information from UE **105** to core network **205** and/or other devices or systems. In some embodiments, some or all of process **700** may be performed by UE **105**. In some embodiments, one or more other devices may perform some or all of process **700** in concert with, and/or in lieu of, UE **105**.

(47) As shown, process **700** may include connecting or requesting (at **702**) connection to a wireless network. For example, UE **105** may connect to base station **101** via a particular band out of a set of different bands implemented by base station **101**. Additionally, or alternatively, UE **105** may output a request to connect to base station **101**, such as via an RRC Connection Request message.

(48) Process **700** may further include providing (at **706**) an indication of the currently connected and/or requested band to core network **205**. For example, as discussed above, UE **105** may communicate with AMF **201** (e.g., via NAS signaling, via an N1 interface, etc.), and may indicate a

particular band **103** to which UE **105** is connected. For example, UE **105** may provide such information periodically or intermittently, based on establishing or re-establishing a connection to base station **101**, and/or on some other basis. In some embodiments, UE **105** may provide the information as part of a request to connect to base station **101** and/or establish one or more communication sessions with core network **205**. In some embodiments, as discussed above, UE **105** may provide (at **706**) the indication based on receiving (at **704**) a request for the band information. For example, as discussed above, UE **105** may receive (at **704**) the request from AMF **201**, via a user input at UE **105**, via a request from an application executing at UE **105**, and/or from some other suitable source.

(49) Process **700** may additionally include receiving (at **706**) service via the wireless network (e.g., via base station **101** and core network **205**) based on the indicated band. For example, as discussed above, one or more elements of core network **205** (e.g., AMF **201**) may select and provide a set of URSP rules to UE **105**, based on which UE **105** may request or establish communication sessions (e.g., PDU sessions) according to particular network slices or other parameters. Additionally, or alternatively, application server **207**, which communicates with UE **105** via the wireless network, may select or adjust one or more service parameters based on the indicated band. As another example, base station **101** and/or one or more other RAN elements may select or adjust parameters based on band information received from UE **105** and/or other UEs (e.g., to reduce or adjust load associated with one or more bands). As shown, some or all of process **700** may be performed and/or repeated iteratively, such that services received by UE **105** may be adjusted in real time or near-real time, such as in situations where UE **105** switches from one band **103** to another.

(50) FIG. **8** illustrates an example environment **800**, in which one or more embodiments may be implemented. In some embodiments, environment **800** may correspond to a 5G network, and/or may include elements of a 5G network. In some embodiments, environment **800** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G RAT may be used in conjunction with one or more other RATs (e.g., a LTE RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”)). In some embodiments, portions of environment **800** may represent or may include a 5G core (“5GC”). As shown, environment **800** may include UE **801**, RAN **810** (which may include one or more Next Generation Node Bs (“gNBs”) **811**), RAN **812** (which may include one or more evolved Node Bs (“eNBs”) **813**), and various network functions such as AMF **201**, MME **816**, Serving Gateway (“SGW”) **817**, Session Management Function (“SMF”)/Packet Data Network (“PDN”) Gateway (“PGW”)–Control plane function (“PGW-C”) **820**, PCF/Policy Charging and Rules Function (“PCRF”) **825**, Application Function (“AF”) **830**, User Plane Function (“UPF”)/PGW–User plane function (“PGW-U”) **835**, Unified Data Management (“UDM”)/Home Subscriber Server (“HSS”) **840**, and Authentication Server Function (“AUSF”) **845**. Environment **800** may also include one or more networks, such as Data Network (“DN”) **850**. Environment **800** may include one or more additional devices or systems communicatively coupled to one or more networks (e.g., DN **850**).

(51) The example shown in FIG. **8** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **820**, PCF/PCRF **825**, UPF/PGW-U **835**, UDM/HSS **840**, and/or AUSF **845**). In practice, environment **800** may include multiple instances of such components or functions. For example, in some embodiments, environment **800** may include multiple “slices” of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of SMF/PGW-C **820**, PCF/PCRF **825**, UPF/PGW-U **835**, UDM/HSS **840**, and/or AUSF **845**, while another slice may include a second instance of SMF/PGW-C **820**, PCF/PCRF **825**, UPF/PGW-U **835**, UDM/HSS **840**, and/or AUSF **845**). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service (“QoS”) parameters.

(52) The quantity of devices and/or networks, illustrated in FIG. **8**, is provided for explanatory

purposes only. In practice, environment **800** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **8**. For example, while not shown, environment **800** may include devices that facilitate or enable communication between various components shown in environment **800**, such as routers, modems, gateways, switches, hubs, etc. Alternatively, or additionally, one or more of the devices of environment **800** may perform one or more network functions described as being performed by another one or more of the devices of environment **800**. Devices of environment **800** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. In some implementations, one or more devices of environment **800** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **800**, such as NEF **303**. For example, some or all elements of the EPC, 5GC, etc. may be communicatively coupled to NEF **303** (e.g., via a Nnef interface or other suitable interface).

(53) UE **801** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **810**, RAN **812**, and/or DN **850**. UE **801** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an Internet of Things (“IoT”) device (e.g., a sensor, a smart home appliance, a wearable device, a Machine-to-Machine (“M2M”) device, or the like), or another type of mobile computation and communication device. UE **801** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **850** via RAN **810**, RAN **812**, and/or UPF/PGW-U **835**.

(54) RAN **810** may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs **811**), via which UE **801** may communicate with one or more other elements of environment **800**. UE **801** may communicate with RAN **810** via an air interface (e.g., as provided by gNB **811**). For instance, RAN **810** may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **801** via the air interface, and may communicate the traffic to UPF/PGW-U **835**, and/or one or more other devices or networks. Similarly, RAN **810** may receive traffic intended for UE **801** (e.g., from UPF/PGW-U **835**, AMF **201**, and/or one or more other devices or networks) and may communicate the traffic to UE **801** via the air interface. In some embodiments, base station **101** may be, may include, and/or may be implemented by one or more gNBs **811**.

(55) RAN **812** may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs **813**), via which UE **801** may communicate with one or more other elements of environment **800**. UE **801** may communicate with RAN **812** via an air interface (e.g., as provided by eNB **813**). For instance, RAN **812** may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **801** via the air interface, and may communicate the traffic to UPF/PGW-U **835**, and/or one or more other devices or networks. Similarly, RAN **812** may receive traffic intended for UE **801** (e.g., from UPF/PGW-U **835**, SGW **817**, and/or one or more other devices or networks) and may communicate the traffic to UE **801** via the air interface. In some embodiments, base station **101** may be, may include, and/or may be implemented by one or more eNBs **813**.

(56) AMF **201** may include one or more devices, systems, Virtualized Network Functions (“VNFs”), Cloud-Native Network Functions (“CNFs”), etc., that perform operations to register UE **801** with the 5G network, to establish bearer channels associated with a session with UE **801**, to hand off UE **801** from the 5G network to another network, to hand off UE **801** from the other network to the 5G network, manage mobility of UE **801** between RANs **810** and/or gNBs **811**, and/or to perform other operations. In some embodiments, the 5G network may include multiple

AMFs **201**, which communicate with each other via the N14 interface (denoted in FIG. **8** by the line marked “N14” originating and terminating at AMF **201**).

(57) MME **816** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **801** with the EPC, to establish bearer channels associated with a session with UE **801**, to hand off UE **801** from the EPC to another network, to hand off UE **801** from another network to the EPC, manage mobility of UE **801** between RANs **812** and/or eNBs **813**, and/or to perform other operations.

(58) SGW **817** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate traffic received from one or more eNBs **813** and send the aggregated traffic to an external network or device via UPF/PGW-U **835**. Additionally, SGW **817** may aggregate traffic received from one or more UPF/PGW-U **835** and may send the aggregated traffic to one or more eNBs **813**. SGW **817** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **810** and **812**).

(59) SMF/PGW-C **820** may include one or more devices, systems, VNFs, CNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **820** may, for example, facilitate the establishment of communication sessions on behalf of UE **801**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **825**.

(60) PCF/PCRF **825** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF **825** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **825**). In some embodiments, PCF **601** may be, may include, and/or may be implemented by PCF/PCRF **825**.

(61) AF **830** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

(62) UPF/PGW-U **835** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **835** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **801**, from DN **850**, and may forward the user plane data toward UE **801** (e.g., via RAN **810**, SMF/PGW-C **820**, and/or one or more other devices). In some embodiments, multiple UPFs **835** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **801** may be coordinated via the N9 interface (e.g., as denoted in FIG. **8** by the line marked “N9” originating and terminating at UPF/PGW-U **835**). Similarly, UPF/PGW-U **835** may receive traffic from UE **801** (e.g., via RAN **810**, SMF/PGW-C **820**, and/or one or more other devices), and may forward the traffic toward DN **850**. In some embodiments, UPF/PGW-U **835** may communicate (e.g., via the N4 interface) with SMF/PGW-C **820**, regarding user plane data processed by UPF/PGW-U **835**.

(63) UDM/HSS **840** and AUSF **845** may include one or more devices, systems, VNFs, CNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **845** and/or UDM/HSS **840**, profile information associated with a subscriber. AUSF **845** and/or UDM/HSS **840** may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE **801**. In some embodiments, UDM **301** may be, may include, and/or may be implemented by UDM/HSS **840**.

(64) DN **850** may include one or more wired and/or wireless networks. For example, DN **850** may include an Internet Protocol (“IP”)-based PDN, a wide area network (“WAN”) such as the Internet, a private enterprise network, and/or one or more other networks. UE **801** may communicate, through DN **850**, with data servers, other UEs **801**, and/or to other servers or applications that are coupled to DN **850**. DN **850** may be connected to one or more other networks, such as a public switched telephone network (“PSTN”), a public land mobile network (“PLMN”), and/or another

network. DN **850** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **801** may communicate.

(65) FIG. **9** illustrates an example RAN environment **900**, which may be included in and/or implemented by one or more RANs (e.g., RAN **810**, RAN **812**, or some other RAN). In some embodiments, a particular RAN may include one RAN environment **900**. In some embodiments, a particular RAN may include multiple RAN environments **900**. In some embodiments, RAN environment **900** may correspond to a particular gNB **811** of a 5G RAN (e.g., RAN **810**). In some embodiments, RAN environment **900** may correspond to multiple gNBs **811**. In some embodiments, RAN environment **900** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, RAN environment **900** may include Central Unit (“CU”) **905**, one or more Distributed Units (“DUs”) **903-1** through **903-N** (referred to individually as “DU **903**,” or collectively as “DUs **903**”), and one or more Radio Units (“RUs”) **901-1** through **901-M** (referred to individually as “RU **901**,” or collectively as “RUs **901**”).

(66) CU **905** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. **8**, such as AMF **201** and/or UPF/PGW-U **835**). In the uplink direction (e.g., for traffic from UEs **801** to a core network), CU **905** may aggregate traffic from DUs **903**, and forward the aggregated traffic to the core network. In some embodiments, CU **905** may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs **903**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol (“PDCP”) packets based on the RLC packets) on the traffic received from DUs **903**.

(67) In accordance with some embodiments, CU **905** may receive downlink traffic (e.g., traffic from the core network) for a particular UE **801**, and may determine which DU(s) **903** should receive the downlink traffic. DU **903** may include one or more devices that transmit traffic between a core network (e.g., via CU **905**) and UE **801** (e.g., via a respective RU **901**). DU **903** may, for example, receive traffic from RU **901** at a first layer (e.g., physical (“PHY”) layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU **903** may receive traffic from CU **905** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **901** for transmission to UE **801**.

(68) RU **901** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **801**, one or more other DUs **903** (e.g., via RUs **901** associated with DUs **903**), and/or any other suitable type of device. In the uplink direction, RU **901** may receive traffic from UE **801** and/or another DU **903** via the RF interface and may provide the traffic to DU **903**. In the downlink direction, RU **901** may receive traffic from DU **903**, and may provide the traffic to UE **801** and/or another DU **903**.

(69) One or more elements of RAN environment **900** may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing (“MEC”) devices, referred to sometimes herein simply as “MECs” **907**. For example, DU **903-1** may be communicatively coupled to MEC **907-1**, DU **903-N** may be communicatively coupled to MEC **907-N**, CU **905** may be communicatively coupled to MEC **907-2**, and so on. MECs **907** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **801**, via a respective RU **901**.

(70) For example, DU **903-1** may route some traffic, from UE **801**, to MEC **907-1** instead of to a core network via CU **905**. MEC **907-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **801** via RU **901-1**. In some embodiments, MEC **907** may include, and/or may implement, some or all of the functionality described above with respect to AF **830**, UPF **835**, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **801**, as traffic

does not need to traverse DU **903**, CU **905**, links between DU **903** and CU **905**, and an intervening backhaul network between RAN environment **900** and the core network.

(71) FIG. **10** illustrates an example O-RAN environment **1000**, which may correspond to RAN **810**, RAN **812**, and/or RAN environment **900**. For example, RAN **810**, RAN **812**, and/or RAN environment **900** may include one or more instances of O-RAN environment **1000**, and/or one or more instances of O-RAN environment **1000** may implement RAN **810**, RAN **812**, RAN environment **900**, and/or some portion thereof. As shown, O-RAN environment **1000** may include Non-Real Time Radio Intelligent Controller (“RIC”) **1001**, Near-Real Time RIC **1003**, O-eNB **1005**, O-CU-Control Plane (“O-CU-CP”) **1007**, O-CU-User Plane (“O-CU-UP”) **1009**, O-DU **1011**, O-RU **1013**, and O-Cloud **1015**. In some embodiments, O-RAN environment **1000** may include additional, fewer, different, and/or differently arranged components.

(72) In some embodiments, some or all of the elements of O-RAN environment **1000** may be implemented by one or more configurable or provisionable resources, such as virtual machines, cloud computing systems, physical servers, and/or other types of configurable or provisionable resources. In some embodiments, some or all of O-RAN environment **1000** may be implemented by, and/or communicatively coupled to, one or more MECs **907**.

(73) Non-Real Time RIC **1001** and Near-Real Time RIC **1003** may receive performance information (and/or other types of information) from one or more sources, and may configure other elements of O-RAN environment **1000** based on such performance or other information. For example, Near-Real Time RIC **1003** may receive performance information, via one or more E2 interfaces, from O-eNB **1005**, O-CU-CP **1007**, and/or O-CU-UP **1009**, and may modify parameters associated with O-eNB **1005**, O-CU-CP **1007**, and/or O-CU-UP **1009** based on such performance information. Similarly, Non-Real Time RIC **1001** may receive performance information associated with O-eNB **1005**, O-CU-CP **1007**, O-CU-UP **1009**, and/or one or more other elements of O-RAN environment **1000** and may utilize machine learning and/or other higher level computing or processing to determine modifications to the configuration of O-eNB **1005**, O-CU-CP **1007**, O-CU-UP **1009**, and/or other elements of O-RAN environment **1000**. In some embodiments, Non-Real Time RIC **1001** may generate machine learning models based on performance information associated with O-RAN environment **1000** or other sources, and may provide such models to Near-Real Time RIC **1003** for implementation.

(74) In some embodiments, Non-Real Time RIC **1001** and/or Near-Real Time RIC **1003** may include, may implement, may be implemented by, and/or may otherwise be associated with RAN management system **203**. In such embodiments, Non-Real Time MC **1001** and/or Near-Real Time MC **1003** may receive band information (e.g., from AMF **201** and/or some other suitable source), and may configure parameters of one or more other elements of O-RAN environment **1000** based on the received band information. For example, as discussed above, Non-Real Time RIC **1001** and/or Near-Real Time RIC **1003** may instruct O-RU **1013** to modify physical parameters of one or more antennas or radios (e.g., azimuth angle, beam width, beam transmit power, etc.) based on band information. Additionally, or alternatively, Non-Real Time RIC **1001** and/or Near-Real Time RIC **1003** may instruct O-DU **1011** and/or O-CU-UP **1009** to prioritize, queue, etc. traffic associated with a given UE **105** in accordance with particular QoS parameters selected based on band information provided by or otherwise associated with UE **105**.

(75) O-eNB **1005** may perform functions similar to those described above with respect to eNB **813**. For example, O-eNB **1005** may facilitate wireless communications between UE **105** and a core network. O-CU-CP **1007** may perform control plane signaling to coordinate the aggregation and/or distribution of traffic via one or more DUs **903**, which may include and/or be implemented by one or more O-DUs **1011**, and O-CU-UP **1009** may perform the aggregation and/or distribution of traffic via such DUs **903** (e.g., O-DUs **1011**). O-DU **1011** may be communicatively coupled to one or more RUs **901**, which may include and/or may be implemented by one or more O-RUs **1013**. In some embodiments, O-Cloud **1015** may include or be implemented by one or more MECs **907**,

which may provide services, and may be communicatively coupled, to O-CU-CP **1007**, O-CU-UP **1009**, O-DU **1011**, and/or O-RU **1013** (e.g., via an O1 and/or O2 interface).

(76) FIG. **11** illustrates example components of device **1100**. One or more of the devices described above may include one or more devices **1100**. Device **1100** may include bus **1110**, processor **1120**, memory **1130**, input component **1140**, output component **1150**, and communication interface **1160**. In another implementation, device **1100** may include additional, fewer, different, or differently arranged components.

(77) Bus **1110** may include one or more communication paths that permit communication among the components of device **1100**. Processor **1120** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. In some embodiments, processor **1120** may be or may include one or more hardware processors. Memory **1130** may include any type of dynamic storage device that may store information and instructions for execution by processor **1120**, and/or any type of non-volatile storage device that may store information for use by processor **1120**.

(78) Input component **1140** may include a mechanism that permits an operator to input information to device **1100** and/or other receives or detects input from a source external to **1140**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **1140** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **1150** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

(79) Communication interface **1160** may include any transceiver-like mechanism that enables device **1100** to communicate with other devices and/or systems. For example, communication interface **1160** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **1160** may include a wireless communication device, such as an infrared (“IR”) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **1100** may include more than one communication interface **1160**. For instance, device **1100** may include an optical interface and an Ethernet interface.

(80) Device **1100** may perform certain operations relating to one or more processes described above. Device **1100** may perform these operations in response to processor **1120** executing software instructions stored in a computer-readable medium, such as memory **1130**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **1130** from another computer-readable medium or from another device. The software instructions stored in memory **1130** may cause processor **1120** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(81) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

(82) For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. **1-6**), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel.

Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

(83) The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

(84) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(85) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

(86) Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

(87) To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

(88) No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A system, comprising: one or more devices of a core of a wireless network; and a User Equipment (“UE”) configured to: connect or request connection to a base station of the wireless

network, wherein the base station implements a plurality of frequency bands, wherein connecting or requesting connection to the base station includes connecting or requesting connection via a particular frequency band of the plurality of frequency bands; and provide an indication of the particular frequency band to the one or more devices of a core of the wireless network, wherein the one or more devices of the core of the wireless network are configured to: maintain information associating a plurality of different sets of UE Route Selection Policy (“URSP”) rules with a plurality of different frequency bands, and select a particular set of URSP rules, out of the plurality of sets of URSP rules, based on identifying that the particular frequency band indicated by the UE is associated with the particular set of URSP rules, wherein the UE is further configured to: receive, from the one or more devices of the core of the wireless network and in response to the provided indication of the particular frequency band, the particular set of URSP rules; and establish one or more communication sessions with the core of the wireless network using the received particular set of URSP rules.

2. The system of claim 1, wherein the one or more devices of the core of the wireless network include at least one of: an Access and Mobility Management Function (“AMF”), or a Mobility Management Entity (“MME”).

3. The system of claim 1, wherein providing the indication of the particular frequency band includes providing the indication via control plane messaging from the UE to the one or more devices of the core of the wireless network.

4. The system of claim 1, wherein providing the indication of the particular frequency band includes providing the indication via Non-Access Stratum (“NAS”) messaging from the UE to the one or more devices of the core of the wireless network.

5. The system of claim 1, wherein providing the indication of the particular frequency band includes providing the indication via an N1 interface.

6. The system of claim 1, wherein the particular set of URSP rules includes information associating a particular set of traffic descriptors with a particular network slice.

7. The system of claim 6, wherein the UE is further configured to: identify that a particular communication session, of the one or more communication sessions, is associated with the particular set of traffic descriptors; and identify, based on the particular set of URSP rules and further based on identifying that the particular communication session is associated with the particular set of traffic descriptors, that the particular communication session is further associated with the particular network slice, wherein establishing the particular communication session includes requesting the particular network slice for the particular communication session.

8. One or more non-transitory computer-readable media, storing a plurality of processor-executable instructions to: connect or request connection between a User Equipment (“UE”) and a base station of a wireless network, wherein the base station implements a plurality of frequency bands, wherein connecting or requesting connection to the base station includes connecting or requesting connection via a particular frequency band of the plurality of frequency bands; provide an indication of the particular frequency band to one or more devices of a core of the wireless network; maintain, by the one or more devices of the core of the wireless network, information associating a plurality of different sets of UE Route Selection Policy (“URSP”) rules with a plurality of different frequency bands; select, by the one or more devices of the core of the wireless network, a particular set of URSP rules, out of the plurality of sets of URSP rules, based on identifying that the particular frequency band indicated by the UE is associated with the particular set of URSP rules; receive, from the one or more devices of the core of the wireless network and in response to the provided indication of the particular frequency band, the particular set of URSP rules; and establish one or more communication sessions with the core of the wireless network using the particular set of URSP rules.

9. The one or more non-transitory computer-readable media of claim 8, wherein the one or more devices of the core of the wireless network include at least one of: an Access and Mobility

Management Function (“AMF”), or a Mobility Management Entity (“MME”).

10. The one or more non-transitory computer-readable media of claim 8, wherein providing the indication of the particular frequency band includes providing the indication via control plane messaging from the UE to the one or more devices of the core of the wireless network.

11. The one or more non-transitory computer-readable media of claim 8, wherein providing the indication of the particular frequency band includes providing the indication via Non-Access Stratum (“NAS”) messaging from the UE to the one or more devices of the core of the wireless network.

12. The one or more non-transitory computer-readable media of claim 8, wherein providing the indication of the particular frequency band includes providing the indication via an N1 interface.

13. The one or more non-transitory computer-readable media of claim 8, wherein the particular set of URSP rules includes information associating a particular set of traffic descriptors with a particular network slice.

14. The one or more non-transitory computer-readable media of claim 13, wherein the plurality of processor-executable instructions further include processor-executable instructions to: identify that a particular communication session, of the one or more communication sessions, is associated with the particular set of traffic descriptors; and identify, based on the particular set of URSP rules and further based on identifying that the particular communication session is associated with the particular set of traffic descriptors, that the particular communication session is further associated with the particular network slice, wherein establishing the particular communication session includes requesting the particular network slice for the particular communication session.

15. A method, comprising: connecting or requesting connection, by a User Equipment (“UE”), to a base station of a wireless network, wherein the base station implements a plurality of frequency bands, wherein connecting or requesting connection to the base station includes connecting or requesting connection via a particular frequency band of the plurality of frequency bands; providing an indication of the particular frequency band to one or more devices of a core of the wireless network; maintaining, by the one or more devices of the core of the wireless network, information associating a plurality of different sets of UE Route Selection Policy (“URSP”) rules with a plurality of different frequency bands; selecting, by the one or more devices of the core of the wireless network, a particular set of URSP rules, out of the plurality of sets of URSP rules, based on identifying that the particular frequency band indicated by the UE is associated with the particular set of URSP rules; receiving, by the UE, from one or more devices of the core of the wireless network and in response to the provided indication of the particular frequency band, the particular set of URSP rules; and establishing one or more communication sessions with the core of the wireless network using the particular set of URSP rules.

16. The method of claim 15, wherein the one or more devices of the core of the wireless network include at least one of: an Access and Mobility Management Function (“AMF”), or a Mobility Management Entity (“MME”).

17. The method of claim 15, wherein providing the indication of the particular frequency band includes providing the indication via control plane messaging from the UE to the one or more devices of the core of the wireless network.

18. The method of claim 15, wherein providing the indication of the particular frequency band includes providing the indication via Non-Access Stratum (“NAS”) messaging from the UE to the one or more devices of the core of the wireless network.

19. The method of claim 15, wherein providing the indication of the particular frequency band includes providing the indication via an N1 interface.

20. The method of claim 15, wherein the particular set of URSP rules includes information associating a particular set of traffic descriptors with a particular network slice, wherein the method further comprises: identifying that a particular communication session, of the one or more communication sessions, is associated with the particular set of traffic descriptors; and identifying,

based on the particular set of URSP rules and further based on identifying that the particular communication session is associated with the particular set of traffic descriptors, that the particular communication session is further associated with the particular network slice, wherein establishing the particular communication session includes requesting the particular network slice for the particular communication session.
